

----- Original Message -----

From: minju.ha@us.nature.com

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Sent: Tue, 21 Jul 2026 04:17:40 -0400

Subject: Decision on Nature Communications manuscript NCOMMS-26-041333-T

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Dear Professor Wei,

Thank you again for submitting your manuscript "Learning the native-like codons with a 5'UTR and RNA secondary structure aided species-informed transformer model" to Nature Communications. We have now received reports from 3 reviewers and, after careful consideration, we have decided to invite a major revision of the manuscript.

As you will see from the reports copied below, the reviewers raise important concerns. We find that these concerns limit the strength of the study, and therefore we ask you to address them with additional work. Without substantial revisions, we will be unlikely to send the paper back to review. We consider it particularly important that the conclusions be supported by additional experimentations that should include additional benchmarking (Reviewer #1, #3) and experimental validation (Reviewer #1, #2).

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Best regards,

Minju Ha, PhD
Senior Editor

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

Hu et al. developed TransCodon, an encoder-only transformer for predicting host-specific synonymous codons. TransCodon uses tokenization at the single-nucleotide level; the 5' UTR is fed into the input alongside the coding sequence (the 5' UTR-CDS boundary is marked by a <sep> token), and a supervised RNA secondary-structure head (ViennaRNA) was added on top of ordinary masked-token training. Species conditioning is done similarly to CodonTransformer, an additive organism embedding broadcast across every token, here via a dedicated learnable table. The model was pretrained on ~5.5M coding sequences across 1,436 bacterial, archaeal, and eukaryotic species, with 5'UTRs included only for the prokaryotic subsets (100 nt upstream of the start). The stated goal is to generate sequences that match the distribution of natural ones rather than to maximize CAI, evaluated on standard codon-usage and structural metrics over a held-out sequence set (based on sequence similarity) from eight species.

-On most of the evaluations, the gain over CodonTransformer is marginal. Codon recovery is essentially tied, and although the sequence is closer to natural codon/GC/folding-energy distributions, a trivial frequency-sampling baseline does about as well.

-The small wet-lab test on six E. coli P450 genes is not very conclusive (better than the conventional tool on two, worse on one, tied on three) and only one other model has been compared to. Using CodonTransformer here as they have taken along many comparisons in the work, would add a fair experiment.

-The pretraining 5' UTR is defined as a fixed 100 nt upstream of the start codon in archaea and bacteria, but in prokaryotes that window routinely captures promoter, operator, RBS, or upstream-gene sequence, and its length and content vary substantially across species, so it isn't a clean UTR. The only UTR benchmark, meanwhile, is the synthetic 50-nt MRL libraries from UTR-LM (originating from Sample et al. 2019, random sequences assayed in human cells), disconnected from the prokaryotic pretraining data in organism, sequence type, and length. And the improvement over UTR-LM is marginal, with the UTR-LM/RNA-FM/RNABERT numbers

taken directly from UTR-LM's paper rather than re-run.

Reviewer #2 (Remarks to the Author):

The manuscript presents TransCodon, a species-informed Transformer model for cross-species codon optimization. The work aims to move beyond conventional CAI- or high-frequency-codon-based optimization by learning host-native coding features, including species-specific codon usage, GC/MFE properties, local %MinMax profiles, low-frequency codon placement, 5'UTR context and RNA secondary-structure information. This is a potentially useful direction, particularly because the manuscript emphasizes native-like codon landscapes rather than simple maximization of frequent codons. However, the central argument is not yet fully supported. The manuscript shows that TransCodon can generate CDSs closer to natural sequence distributions, but it does not sufficiently demonstrate that such native-likeness reliably leads to improved heterologous expression, solubility or protein function. In particular, the practical role of 5'UTR information during inference and in the P450 validation remains unclear, although this is a major claim in the title and framing of the study. While I believe this study holds some value, but substantial amount of work is needed to substantiate some of the claims in this work.

Major comments

- 1.The relationship between native-like CDS generation and protein-yield optimization needs to be clarified. The manuscript repeatedly frames TransCodon as a codon-optimization method for improved heterologous expression. For example, the Abstract on page 1 states that the P450 validation demonstrated superior expression outcomes compared with DNAworks and that the model may generate native-like CDSs with high translational efficiency. Similar claims appear in the Introduction on page 2 and the Conclusion on page 15. However, on page 11, the authors state that TransCodon is designed to match host-native codon usage bias and is not specifically tailored to directly maximize protein yield. This creates a conceptual tension. Native-likeness should not be conflated with expression optimization. The authors should clearly define what TransCodon optimizes and temper expression-related claims accordingly.
- 2.The practical use of 5'UTR-guided codon optimization is underdefined. The title, Abstract and Figure 1 emphasize 5'UTR information, and the Methods section on page 16 defines 5'UTRs as the 100 nucleotides upstream of the start codon in Datasets 2 and 3.
3. However, in heterologous expression, there is no unique 5'UTR to provide during inference. It could come from the expression vector, the target host, the native gene, a highly expressed host gene or a synthetic design. In addition, because the model was

trained using a fixed 100-nt upstream window, it is unclear whether inference should use the full 5'UTR, or only the 100 nt immediately upstream of the start codon. Using full-length 5'UTRs at inference could introduce a train-test distribution mismatch, whereas using fixed 100-nt windows requires a clear biological and practical rationale. The manuscript does not specify how users should choose or truncate a 5'UTR, nor does it test how different 5'UTR inputs affect generated CDSs or expression outcomes. Thus, the current results show that 5'UTR information may help representation learning, but not yet that 5'UTR-conditioned generation improves CDS design for expression.

3. The manuscript should more clearly distinguish CDS-only modeling from 5'UTR-conditioned modeling. A substantial part of the manuscript, including the main benchmarks in Figures 2-5, appears to evaluate species from Dataset1, which Table 1 indicates does not contain 5'UTR information. These results therefore mainly demonstrate that TransCodon can learn CDS-level codon usage patterns and generate more native-like CDSs, rather than demonstrating the added value of 5'UTR information. This is not fully aligned with the manuscript's central framing, which presents 5'UTR integration as a major contribution. In contrast, Supplementary Figure S2 is the main analysis that directly tests 5'UTR involvement, and it suggests that 5'UTR information is beneficial only when it is provided during both training and inference; omitting it at inference can reduce performance. The authors should explicitly state which experiments used 5'UTR inputs, which were CDS-only, and which claims are supported by each setting. Ideally, they should provide matched comparisons between CDS-only and 5'UTR-conditioned models on the same species and test sets.

4. The P450 validation does not specify whether 5'UTR information was used during TransCodon inference. The main text on pages 10-11 and Figure 7 state that six P450 protein sequences were designed using TransCodon or DNAworks and expressed in *E. coli*. The Supplementary Information, Part 6 and Table S2, specify the host and vector, but do not report whether any 5'UTR was provided to TransCodon during inference. If a 5'UTR was used, its source, sequence, length and relationship to the pET-21a expression context should be reported. If no 5'UTR was used, the wet-lab validation does not directly support the claimed role of 5'UTR-guided CDS optimization.

5. The P450 validation is too small to support the current strength of the expression claims. Figure 7 and Supplementary Table S4 show that, at 37 °C, TransCodon significantly outperformed DNAworks for P450-4 and P450-5, was significantly worse for P450-3, and showed no significant difference for P450-1, P450-2 and P450-6. This supports a more limited conclusion that TransCodon can be beneficial in some cases, but not that it generally improves heterologous expression. In addition, the reported correlations between model predictions and experimental yields across these six proteins (Spearman 0.486, Pearson 0.525 for the TransCodon fitness score; Spearman

0.886, Pearson 0.795 after adding an MLP abundance-prediction head) should be interpreted cautiously. With only six data points, these correlations are highly sensitive to individual proteins and do not establish that the model can prospectively screen high-expression sequences before experiments. Confidence intervals or permutation/bootstrap analyses would help quantify this uncertainty. The authors should temper claims such as “superior expression outcomes” and “generally outperformed”, or expand the validation across more proteins, hosts and expression systems.

6.The description of training dataset sizes is internally inconsistent. The Figure 2 legend on page 5 and the Supplementary Figure S1 legend state that the training sets were 1M (Dataset1), 2.2M (Dataset1+Dataset2), 4M (Dataset1+Dataset2+Dataset3), 5.5M (Dataset1+Dataset2+Dataset4), and 11M (Dataset1+Dataset2 plus 8.8M additional bacterial sequences). However, Table 1 on page 16 lists Dataset1, Dataset2, Dataset3 and Dataset4 as approximately 1.0M, 1.4M, 1.7M and 1.4M sequences, respectively. Thus, Dataset1+Dataset2 should be about 2.4M, and Dataset1+Dataset2+Dataset4 should be about 3.8M rather than 5.5M. The 5.5M setting appears to require all four datasets, and the extra 8.8M bacterial sequences are not described in the Methods table. This should be corrected and clarified.

7.The synonymous-codon-constrained decoding procedure should be described explicitly. Because TransCodon uses nucleotide-level tokenization and masked nucleotide prediction, it is essential to explain how inference preserves the target amino-acid sequence. The Methods section on page 17 gives the example MKD encoded as ATGAAGA, but this does not explain amino acids such as Leu, Ser and Arg, whose synonymous codons differ at the first or second nucleotide. The public code appears to address this by enumerating all synonymous codons for each amino acid and selecting the codon with the highest joint nucleotide probability, but this important step is not described in the manuscript. The authors should report this constrained decoding algorithm and provide an amino-acid preservation rate for generated CDSs.

8.The 5'UTR downstream evaluation is limited. Section 2.4 on page 10 and Figure 6 focus on MRL prediction, and the Figure 6 legend states that RNABERT, RNA-FM and UTR-LM results were taken directly from the original UTR-LM publication. Given that 5'UTR integration is presented as a major contribution, one MRL task with partly non-reproduced baselines is not sufficient. The authors should evaluate under a unified protocol against stronger RNA/mRNA foundation-model baselines and consider broader benchmarks, such as 5'UTR tasks in mRNABench (<https://openreview.net/forum?id=JcFz2WkCfy>).

Minor comments

9.The manuscript treats the 100 nucleotides upstream of the start codon as 5'UTR, but

this operational definition is not necessarily biologically justified. The authors should clarify whether these regions are annotated 5'UTRs or fixed upstream windows, and assess whether the conclusions are robust to different upstream lengths.

10. Figures 2, 5 and 6 rely heavily on averaged bar plots. Distribution-aware plots, individual points and confidence intervals would better show sequence-level, species-level and replicate-level variability.

11. The low-frequency-codon analysis in Figure 8 is interesting, but the interpretation should be cautious. Preserving low-frequency codon patterns does not by itself prove improved expression.

12. The Conclusion on page 15 uses strong language such as transformative paradigm and unprecedented biological fidelity. These claims should be softened.

13. Statistical reporting for the P450 experiment should be strengthened, including replicate definition, normalization, multiple testing and assumptions behind Student's t-tests.

Code and reproducibility comments

14. The released code is difficult to audit because it contains many unused components, commented-out branches, hard-coded paths, debugging print statements and duplicated code. The authors should clean the repository and clearly document which scripts reproduce the reported results.

15. `finetune.py` line 227 contains a syntax error (`args.train_data,, args.batch_size`) and cannot be run as released.

16. `infer.py` treats 5'UTR as an optional user-provided input through `--with_utr`, reinforcing the need to define how 5'UTRs should be selected for practical use.

17. The released data-loading pipeline in `transcodon/dataset.py` appears inconsistent with the method described in the manuscript. If the model is trained using the current code path, 5'UTR sequences and RNA secondary-structure labels would not be loaded, which would contradict the claimed joint CDS, 5'UTR and RNA-structure training.

Specifically, `transcodon/dataset.py` lines 65-67 place 5'UTR loading under an `if False` branch and reset `utr_sequence=""`, while lines 80-83 place RNA secondary-structure loading under another `if False` branch and set `struct_label_2d_file='*'`; The authors should clarify whether the released repository corresponds to the exact code used for the reported models and, if not, provide the actual training code and input data schema.

18. The synonymous-codon decoding algorithm implemented in the inference code should be documented in the Methods and README, together with validation that generated CDSs translate to the intended proteins.

Reviewer #3 (Remarks to the Author):

The paper by Hu et al. introduces TransCodon, a BERT-like model of protein coding DNA. The authors train a standard 12-layer model, and show that it can be used in a multitude of tasks, including generating sequences with the right codon distribution across species, and predicting some properties such as mean protein abundance. The most valuable contribution of the paper, is the demonstration that their model could be used to optimise the heterologous expression of some P450 enzymes.

The manuscript is technically sound and well-written. While the contribution in terms of machine learning innovations is relatively limited (the architecture is very similar to ESM1 family of models published in 2021, and a good fraction of the problems have been extensively studied elsewhere), I would argue that the fundamental problem in codon optimization models has always been the lack of validation. One thing is to demonstrate that the sequences look like natural sequences, and a very different one is to show that you can take a protein sequence and efficiently express it in a heterologous system. In my opinion, this wet lab validation is enough that the manuscript should be considered for publication.

The questions I would pose the authors before the article is ready for publication are:

- I am confused by the inference method employed by the authors: in the current writeup, it looks like the authors are using the model to infer the identity of the wobble base, but it is unclear how this would work e.g. in leucine that has non-wobble degeneration (e.g. CUN as well as UUA/UUG encode for Leu). It is also unclear if all the codons are masked at once, or if they use some sort of evolutionary process.
- To follow from the above, I am curious how the generation process changes with different inference strategies, for example if the authors use an "evolution-like" approach to iteratively refine the sequence, similar to e.g. the ESM3 paper.
- The baselines chosen by the authors in computational benchmarks are somewhat limited: they are using just a baseline model (CodonTransformer), and trivial statistical metrics. However, the tokenization used for the models means that they could be directly comparable with other nucleotide models e.g. NucleotideTransformer or Evo/Evo-2. I would like to see how these models compare with the authors' in terms of the benchmarks they generated. This is particularly interesting in the context of zero-shot protein abundance, which was also reported by e.g. Evo. Some of these models are specific to some species (e.g. NucleotideTransformer is human-centric, Evo is prokaryotic), which will make the comparison more interesting.

- The data presented in Figure 7-b is intriguing: yes, two of the sequences are expressed with significantly higher yield (P450-4, P450-5), but another one is expressed at less than half the yield (P450-3)! Could the authors provide an analysis of why there was such a difference? Is it due to the presence of rare codons somewhere important?

Some other minor points:

- Why did the authors select P450 as a test system? One of the questions is obviously whether this system is representative of a general protein yield optimization experiment, or whether there are other factors in play.
- The authors provided broad information about their datasets in the Methods section, but I would like to see more information on how Datasets 2-4 were constructed, and in particular how they selected the specific species used, and defined the 5'-UTR regions

Carlos Outeiral

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